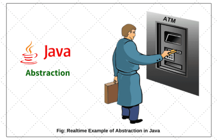


Fig: Realtime Example of Abstraction in Java

ABSTRACTION

Abstraction in Java is a concept that helps in hiding the complex reality while exposing only the necessary features of an object. In Java, abstraction is achieved using abstract classes and interfaces.

Fig: Realtime Example of Abstraction in Java

Abstract Classes

An abstract class in Java is a class that cannot be instantiated on its own and is meant to be subclassed. It may contain abstract methods as well as concrete (fully defined) methods. Abstract classes are used to represent generic concepts or base behaviors that subclasses should implement or expand upon.

Characteristics of Abstract Classes:

- Cannot be instantiated: You cannot create objects of an abstract class directly.
- Can have constructors: Abstract classes can have constructors, and the constructor is called when a subclass is instantiated.
- Can have instance variables: Abstract classes can declare fields that are not static and final, and define methods that can access and modify the state of the object to which they belong.
- Can have defined methods: Abstract classes can have fully implemented methods with actual code.
- Can have abstract methods: Methods without a body, just declared with the keyword abstract and a semicolon at the end.

Abstract Methods

Abstract methods are methods **declared in an abstract class that do not have an implementation**. Subclasses that extend an abstract class must provide implementations for its abstract methods unless the subclass is also abstract.

Characteristics of Abstract Methods:

•No body: Abstract methods do not have a body; they're just a signature followed by a semicolon.

•Implementation by subclasses: Any concrete subclass must implement all inherited abstract methods, or it too must be declared abstract.

•Enforces a contract: Declaring an abstract method in a class imposes a contract on the subclasses, obliging them to support particular behaviors.

The Purpose of Abstraction:

- To control the complexity of large software systems by hiding the implementation details and exposing only the necessary parts of an object.
- To provide a common template for subclasses that can be adapted and implemented as per the subclass's specific requirements.
- To enable code reuse by defining non-specific method signatures in the parent class that different child classes can share and implement in different ways.

```
abstract class Shape {
   abstract double area();   }
class Triangle extends Shape {
   double base, height;
   Triangle(double base, double height) {
      this.base = base;
      this.height = height;  }
   @Override
   double area() {
      return 0.5 * base * height;   } }
class Rectangle extends Shape {
   double length, width;
   Rectangle(double length, double width) {
      this.length = length;
      this.width = width;   }
   @Override
   double area() {
      return length * width;  }}
public class GeometryTest {
   public static void main(String[] args) {
      Shape triangle = new Triangle(3, 4);
      Shape rectangle = new Rectangle(5, 6);
      System.out.println("Triangle area: " + triangle.area());
      System.out.println("Rectangle area: " + rectangle.area());   }}
```

```java
// Declares an abstract class named Shape. An abstract class cannot be instantiated.
abstract class Shape {
   // Declares an abstract method named area. It has no body and must be overridden in subclasses.
   abstract double area();
}
// Declares a concrete class named Triangle that extends the abstract Shape class.
class Triangle extends Shape {
   // Instance variables to store the dimensions of the triangle.
   double base, height;
   // Constructor for the Triangle class. It initializes the base and height of the triangle.
   Triangle(double base, double height) {
      this.base = base;
      this.height = height;
   }
   // Overrides the abstract area method from the Shape class. Calculates and returns the area of the triangle.
   @Override
   double area() {
      return 0.5 * base * height;
   }
}
// Declares a concrete class named Rectangle that extends the abstract Shape class.
class Rectangle extends Shape {
   // Instance variables to store the dimensions of the rectangle.
   double length, width;

   // Constructor for the Rectangle class. It initializes the length and width of the rectangle.
   Rectangle(double length, double width) {
      this.length = length;
      this.width = width;
   }
   // Overrides the abstract area method from the Shape class. Calculates and returns the area of the rectangle.
   @Override
   double area() {
      return length * width;
   }
}
// Contains the main method, which is the entry point of the program.
public class GeometryTest {
   public static void main(String[] args) {
      // Creates a Triangle object with base 3 and height 4, referenced by a Shape variable.
      Shape triangle = new Triangle(3, 4);
      // Creates a Rectangle object with length 5 and width 6, referenced by a Shape variable.
      Shape rectangle = new Rectangle(5, 6);
      // Calls the area method on the triangle object and prints the result.
      System.out.println("Triangle area: " + triangle.area());
      // Calls the area method on the rectangle object and prints the result.
      System.out.println("Rectangle area: " + rectangle.area());
   }
}
```

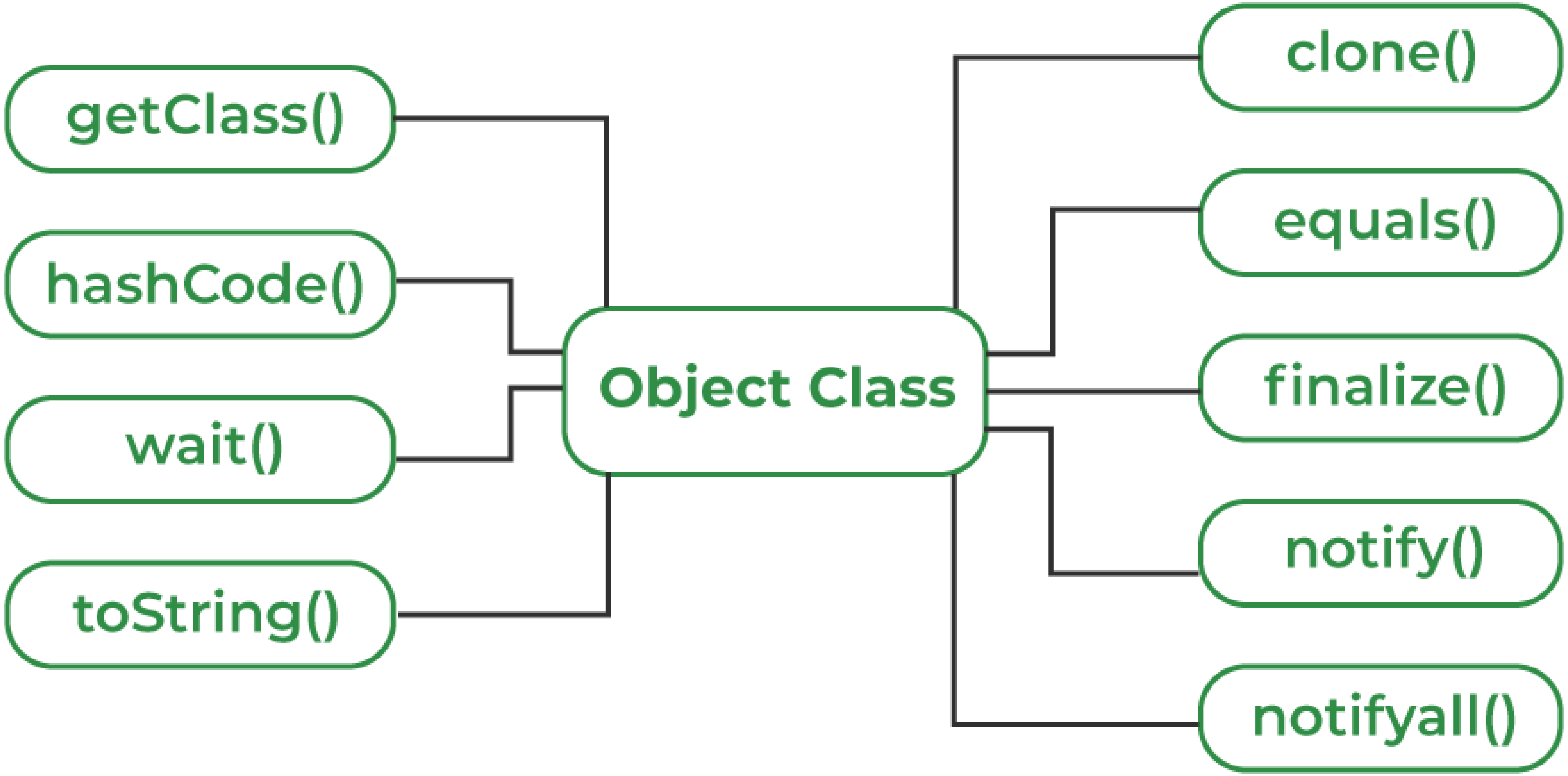
```java
// Define an abstract class for a generic payment processor.
abstract class PaymentProcessor {
  // Abstract method that needs to be implemented by subclasses to process payments.
  abstract void processPayment(double amount);
}

// Subclass of PaymentProcessor for processing credit card payments.
class CreditCardProcessor extends PaymentProcessor {
  // Implement the processPayment method for credit card processing.
  @Override
  void processPayment(double amount) {
    System.out.println("Processing credit card payment of $" + amount);
  }
}
// Subclass of PaymentProcessor for processing payments through PayPal.
class PayPalProcessor extends PaymentProcessor {
  // Implement the processPayment method for PayPal processing.
  @Override
  void processPayment(double amount) {
    System.out.println("Processing PayPal payment of $" + amount);
  }
}
// Main class to test the payment processing subclasses.
public class PaymentTest {
  public static void main(String[] args) {
    // Create an instance of CreditCardProcessor.
    PaymentProcessor creditCard = new CreditCardProcessor();
    // Create an instance of PayPalProcessor.
    PaymentProcessor payPal = new PayPalProcessor();

    // Process a payment using the credit card processor.
    creditCard.processPayment(100.0);
    // Process a payment using the PayPal processor.
    payPal.processPayment(200.0);
  }
}
```

Object Class

Object class in Java

- The **Object class** is the parent class of all the classes in java by default. In other words, it is the topmost class of java.
- The Object class is beneficial if you want to refer any object whose type you don't know. Notice that parent class reference variable can refer the child class object, know as upcasting.
- Let's take an example, there is getObject() method that returns an object but it can be of any type like Employee,Student etc, we can use Object class reference to refer that object

- clone(): Creates and returns a copy of this object.
- equals(Object obj): Indicates whether some other object is "equal to" this one.
- finalize(): Called by the garbage collector on an object when garbage collection determines that there are no more references to the object.
- getClass(): Gets the runtime class of this Object.
- hashCode(): Returns a hash code value for the object.
- toString(): Returns a string representation of the object.
- notify(), notifyAll(), and wait(): Are part of the mechanism that Java uses for thread synchronization.


getClass()
hashCode()
wait()
toString()
Object Class
clone()
equals()
finalize()
notify()
notifyall()

- toString()

```
public class Car {
   private String make;
   private String model;
   public Car(String make, String model) {
      this.make = make;
      this.model = model;
   }
   // Override toString() method from Object class
   @Override
   public String toString() {
      return "Car{" +
            "make='" + make + '\'' +
            ", model='" + model + '\'' +
            '}';
   }
   public static void main(String[] args) {
      Car myCar = new Car("Toyota", "Corolla");
      System.out.println(myCar.toString());  // Explicitly calling toString()
      System.out.println(myCar);            // Implicitly calling toString()
   } }
```

O/P: Car{make='Toyota', model='Corolla'}

```java
class Coordinate {
    int x, y;
    Coordinate(int x, int y) {
        this.x = x;
        this.y = y;
    }
    // Override toString method
    @Override
    public String toString() {
        return "Coordinate: [" + x + ", " + y + "]";
    }
    public static void main(String[] args) {
        Coordinate point = new Coordinate(5, 10);
        System.out.println(point);
    }
}
```

O/P : Coordinate: [5, 10]

```java
class Box {
    int width, height, depth;
    // Constructor for Box class
    Box(int w, int h, int d) {
        width = w;
        height = h;
        depth = d;
    }
    // Overriding the toString method
    @Override
    public String toString() {
        return "Box(" + width + ", " + height + ", " + depth + ")";
    }
    public static void main(String[] args) {
        Box myBox = new Box(10, 20, 30);
        System.out.println(myBox);
    }
}
```

```
// Here we have a class Animal, which is like a general type of object.
class Animal {
  void whoAmI() {
    System.out.println("I am an animal");
  } }
// Here is a more specific type of Animal, let's say a Dog.
class Dog extends Animal {
  void whoAmI() {
    System.out.println("I am a dog");
  }}
public class Main {
  public static void main(String[] args) {
    // Here we create a Dog, but we refer to it as an Animal.
    // This is upcasting. We're treating the Dog just as a general Animal.
    Animal myDog = new Dog();
    // When we call the whoAmI method, it says "I am a dog" because
    // even though we're treating it as an Animal, it's actually a Dog.
    myDog.whoAmI();
  } }
```

UPCASTING

Upcasting is the process of treating a subclass object as an instance of its superclass

```java
class Fruit {
  void show() {
    System.out.println("This is a fruit.");
  } }
class Apple extends Fruit {
  void show() {
    System.out.println("This is an apple.");
  } }
class Banana extends Fruit {
  void show() {
    System.out.println("This is a banana.");
  } }
public class Main {
  public static void main(String[] args) {
    Object fruitBox1 = new Apple(); // Apple is treated as an Object
    Object fruitBox2 = new Banana(); // Banana is treated as an Object
    // Despite being specific fruits, we refer to them using the Object class
    System.out.println(fruitBox1.getClass().getName()); // Outputs "Apple"
    System.out.println(fruitBox2.getClass().getName()); // Outputs "Banana"
  } }
```

- getClass()

```
public class GetClassExample {

  public static void main(String[] args) {
    // Create an instance of String
    String myString = "Hello, World!";

    // Use getClass() to print out the class of the object
    System.out.println("The class of myString is: " +
myString.getClass().getName());
  }
}
```

The class of myString is: java.lang.String

Inner Class in Java

Java

Java Inner Classes (Nested Classes)

Java inner class or nested class is a class that is declared inside the class or interface.

We use inner classes to logically group classes and interfaces in one place to be more readable and maintainable.

Additionally, it can access all the members of the outer class, including private data members and methods.

Syntax of Inner class

```
class Java_Outer_class
{
 //code
 class Java_Inner_class
{
  //code
 }  }
```

- ❑ **Non-static nested class (inner class)**
  - **Member inner class**
  - **Anonymous inner class**
  - **Local inner class**

- ❑ **Static nested class**

Member Inner Class: A class created within class and outside method.

```
class OuterClass
 {
   class InnerClass
      {
      // methods and fields - can access members of the OuterClass.
      }
}
```

```java
public class Circle {
    private double radius;
    private Point center; // Instance of the inner class
    public Circle(double radius, double x, double y) {
        this.radius = radius;
        this.center = new Point(x, y);
    }
    public void displayCenter() {
        System.out.println("Circle center at (" + center.x + ", " + center.y + ")");
    }
    // Member Inner Class
    class Point {
        double x, y;
        public Point(double x, double y) {
            this.x = x;
            this.y = y;
        }
    }
    public static void main(String[] args) {
        Circle circle = new Circle(5.0, 1.0, 2.0);
        circle.displayCenter();
    }
}
```

Local Inner Class: A class defined within a block, typically a method block.

```
class OuterClass
 {
   void myMethod()
   {
      class LocalInnerClass
     {
        // can only be used within the block where it is defined.
      }
   }
}
```

```java
public class Greeting {
    public void greetInLanguage(String language) {
        // Local Inner Class
        class Greeter {
            void greet() {
                switch (language) {
                    case "English":
                        System.out.println("Hello!");
                        break;
                    case "Spanish":
                        System.out.println("Hola!");
                        break;
                    case "French":
                        System.out.println("Bonjour!");
                        break;
                    default:
                        System.out.println("Language not supported.");
                }
            }    }
        Greeter greeter = new Greeter();
        greeter.greet();
    }
    public static void main(String[] args) {
        Greeting greeting = new Greeting();
        greeting.greetInLanguage("French");} }
```

Static Nested Class: A static class created inside a class.

```
class OuterClass
 {
   static class NestedStaticClass
 {
      // can exist independently of an instance of
OuterClass
}
}
```

```
public class Computer {
   private String name;
   private Processor processor;
   public Computer(String name, String processorType, double processorSpeed) {
      this.name = name;
      this.processor = new Processor(processorType, processorSpeed);
   }
   public void displayComputerDetails() {
      System.out.println("Computer Name: " + name);
      System.out.println("Processor Type: " + processor.getType());
      System.out.println("Processor Speed: " + processor.getSpeed() + "GHz");
   }
   // Static Nested Class
   static class Processor {
      private String type;
      private double speed;
      public Processor(String type, double speed) {
         this.type = type;
         this.speed = speAed;
      }
      public String getType() {
         return type;
      }
      public double getSpeed() {
         return speed;
      }
   }
   public static void main(String[] args)
 {
      Computer myComputer = new Computer("MyComputer", "Intel i7", 3.5);
      myComputer.displayComputerDetails();
   }
```


Java Garbage Collection

Java Garbage Collection

- In java, garbage means unreferenced objects.
- Garbage Collection is process of reclaiming the runtime unused memory automatically.
- In other words, it is a way to destroy the unused objects.
- To do so, we were using free() function in C language and delete() in C++. But, in java it is performed automatically. So, java provides better memory management.

How can an object be unreferenced?

01 By nulling the reference

02 By assigning a reference to another

03 By anonymous object etc.

Nulling the Reference

When you set a reference to null, you are explicitly stating that the reference no longer points to any object. After this point, the object it previously pointed to is eligible for garbage collection if there are no other references to it.

```
public class Example
{
   public static void main(String[] args)
 {
      String str = new String("Example string");
      str = null; // Now the String object is eligible for garbage
collection.
   }
}
```

Assigning a Reference to Another

If you have a reference variable that points to an object, and you assign it to another object, the first object becomes unreferenced if no other references exist.

```
public class Example
{
   public static void main(String[] args)
 {
      StringBuilder builder = new StringBuilder("Initial");
      StringBuilder anotherBuilder = new StringBuilder("Another");

      builder = anotherBuilder; // The StringBuilder object containing
"Initial" is now eligible for garbage collection.
   }
}
```

Anonymous Object

An anonymous object is an object that is instantiated but not stored in a reference variable. After the line of code that creates and uses the anonymous object is executed, the object is eligible for garbage collection because it cannot be reached by any live threads.

```
public class Example {
   public static void main(String[] args) {
      new String("This is an anonymous object"); // After this statement,
the String object is eligible for garbage collection.

      // Here we use an anonymous object to perform an operation.
      System.out.println(new StringBuilder("Anonymous
StringBuilder").reverse());
   }
}
```

gc() method

The gc() method is used to invoke the garbage collector to perform cleanup processing. The gc() is found in System and Runtime classes.

```
1.public static void gc(){}
```

Finalize Method.

Java Object finalize() Method

Finalize() is the method of Object class. This method is called just before an object is garbage collected. finalize() method overrides to dispose system resources, perform clean-up activities and minimize memory leaks.

finalize() method

The finalize() method is invoked each time before the object is garbage collected. This method can be used to perform cleanup processing. This method is defined in Object class as:

```
protected void finalize(){}
```

```
public class TestGarbage1{
 public void finalize()
{ System.out.println("object is garbage collected");}
 public static void main(String args[]){
  TestGarbage1 s1=new TestGarbage1();
  TestGarbage1 s2=new TestGarbage1();
  s1=null;
  s2=null;
  System.gc();
 }
}
```

```
class FinalizationDemo {

   @Override
   protected void finalize() {
      System.out.println("Finalize method called.");
   }

   public static void main(String[] args) {
      FinalizationDemo obj = new FinalizationDemo();
      obj = null; // Make obj eligible for garbage collection

      // Suggesting garbage collection. Note that it is NOT guaranteed to run
immediately.
      System.gc();
      System.out.println("Garbage Collection is requested");
   }
}
```